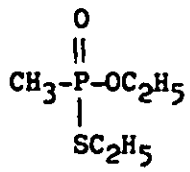
MILITARY SPECIFICATION

O,S-DIETHYL METHYLPHOSPHONOTHIOLATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of O,S-diethyl methylphosphonothiolate, hereinafter referred to as OSDMP, having the following chemical structure:



2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171-179.")

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-70 - Reagent Water.
E191-64 - Apparatus for Microdetermination of Carbon and Hydrogen in Organic and Organo-Metallic Compounds.
E357-68 - Phosphorus in Organic Compounds by Peroxide Bomb Ignition.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. OSDMP shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight	Test paragraph
Purity (as O,S-diethyl methylphosphonothiolate), minimum	90.0	4.2.4.1
Elemental carbon	35.7 ± 0.3	4.2.4.2
Elemental hydrogen	7.80 ± 0.30	4.2.4.2
Elemental phosphorus	18.41 ± 0.30	4.2.4.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his

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own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the OSDMP produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling for test. A representative specimen of approximately 10 grams (g) shall be removed from each lot of OSDMP offered for acceptance and placed in a clean, dry, closed container labeled to identify the lot represented.

4.2.3 Inspection procedure for test. The specimen taken as specified in 4.2.2 shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Tests shall be performed in a well-ventilated hood. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.2.4.1 Purity.

(a) Sodium ethoxide. Dissolve 4 g of freshly cut sodium metal in 100 milliliters (ml) of absolute ethanol in a flask under a water condenser. Cool the flask in an ice bath if the reaction becomes too violent. Keep the sodium ethoxide container cold in a Dewar flask with ice until ready for use. Allow the sodium ethoxide to warm to room temperature just before using. Any orange-brown coloration which may develop will not obscure subsequent end point determinations.

(b) Procedure, part I. Weigh to the nearest 0.1 milligram (mg) approximately 0.2 to 0.4 g of the specimen into a glass ampule. Seal the ampule and place it in a 500-ml iodine flask containing 25 ml of sodium ethoxide prepared as specified in (a) under a continuous dry nitrogen sparge. Under a blanket of dry nitrogen, break the ampule and stem with a glass rod and wash down the rod with no more than 5 ml

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of absolute ethanol. Stopper the flask, sealing it with absolute ethanol. Immediately swirl the flask. Allow the flask to stand for 25 minutes. Wash down the stopper and flask with 100 ml of water and add 100 ml of crushed ice. Add 25 ml of glacial acetic acid and adjust the volume in the flask to 300 to 350 ml (pH 3.5 to 4.0). Add 5 g of potassium iodide, dissolve, and titrate with approximately 0.1N iodine solution to the first faint yellow color.

(c) Procedure, part II. Weigh to the nearest 0.1 mg approximately 0.5 g of the specimen into a glass ampule. Seal the ampule and place it in a 500-ml iodine flask containing 25 ml of glacial acetic acid. Break the ampule and stem with a glass rod. Wash down the rod with water and then dilute the solution to approximately 300 ml with water. Add 5 g of potassium iodide, dissolve, and titrate with approximately 0.1N iodine solution from a 10-ml microburet to the first permanent faint yellow color.

(d) Calculation. Calculate the percent by weight purity as O,S-diethyl methylphosphonothiolate as follows:

$$\text{Percent purity} = 16.82\left(\frac{AB}{C} - \frac{DE}{F}\right)$$

where: A = Milliliters of iodine solution used in the part I titration,
B = Normality of iodine solution used in the part I titration,
C = Weight of specimen in grams in part I,
D = Milliliters of iodine solution used in the part II titration,
E = Normality of iodine solution used in the part II titration, and
F = Weight of specimen in grams in part II.

4.2.4.2 Elemental carbon and elemental hydrogen. Determine the percent by weight of elemental carbon and the percent by weight of elemental hydrogen in the specimen in accordance with ASTM E191.

4.2.4.3 Elemental phosphorus. Determine the percent by weight of elemental phosphorus in the specimen in accordance with ASTM E357.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. A quantity of OSDMP as specified in the contract or order (see 6.2) shall be packaged and packed as a Department of Transportation class B poison in containers which comply with the Code of Federal Regulations and the Uniform Freight Classification rules or the National Motor Freight Classification rules, as applicable. Containers shall assure safe delivery to destination without damage to contents.

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5.2 Marking. Containers shall be marked in accordance with the Code of Federal Regulations and shall also show the following information:

(a) Nomenclature
(b) Net weight of contents
(c) Lot number
(d) Name of manufacturer
(e) Date of manufacture
(f) Percent purity

6. NOTES

6.1 Intended use. OSDMP is intended for use in testing the M8 Automatic Chemical Agent Alarm.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Preparation of OSDMP. A method of preparing and purifying OSDMP is described in the appendix to this specification.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.6 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-D-1094, dated 2 September 1971.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6810-A985

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APPENDIX

PREPARATION AND PURIFICATION OF OSDMP

10. SCOPE

10.1 This appendix describes a method for the preparation and purification of OSDMP. This procedure is furnished with the distinct understandings:

(a) That the supplier is not required to follow the method.

(b) That the requirements of the specification must be met regardless of the method of manufacture.

(c) That the Government makes no warranty that OSDMP made in accordance with this method will meet the requirements of the specification.

20. PROCEDURE

20.1 Preparation of dicyclohexylamine salt of ethyl methylphosphonothioic acid. Place 100 g of diethyl methylphosphonite into a 1-liter flask fitted with a mechanical stirrer, a self-compensating dropping funnel, an immersion thermometer, and an efficient water condenser which is terminated with a drying tube containg "Drierite" or equal. While stirring the liquid and using external cooling, add 13.2 g of water dropwise. Allow the reaction mixture to stand at ambient temperature for 2 hours. Cool the mixture to approximately 5° C, stir, add 400 ml of dry benzene and 133 g of freshly distilled dicyclohexylamine, and stir again. Add 23.6 g of sulfur using intermittent cooling so as to maintain a reaction temperature of 40° C. After the addition of the sulfur is completed, bring the reaction mixture temperature to 26° C and the flask pressure to 90 millimeters (mm) of mercury and remove enough benzene and ethanol to cause the precipitation of the dicyclohexyl salt of ethyl methylphosphonothioic acid. Collect the salt on a Buchner funnel, wash with petroleum ether, and dry the salt in a desiccator over phosphorus pentoxide. Approximately 170 g of salt (75 percent yield) having a melting point of 159° to 161° C will be produced. An additional 56 g of salt having a melting point of 154° to 156° C can be obtained by concentrating the mother liquid. The total salt yield will be approximately 95.8 percent.

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20.2 Preparation of OSDMP. The preparation of OSDMP should be conducted in a well-ventilated hood. Place 125 g of the dicyclohexylamine salt prepared as described in 20.1 in a 1-liter, three-neck flask fitted with the same equipment as the flask in 20.1 and containing 400 ml of anhydrous benzene. Stir the mixture and heat at 50° C. After all the salt has dissolved, add 72 g of ethyl iodide dropwise. Heat under a gentle reflux for 2 hours. Filter the mixture through a Buchner funnel to remove the solid by-product and wash the solid with four 125-ml portions of benzene. Combine the filtrate and washings and remove the solvent at a temperature of 24° C and a pressure of 90 mm of mercury. Distill the oily residue using a 12-inch glass helix packed column at a boiling point of 46° to 47° C and a pressure of 0.5 mm of mercury. The yield will be approximately 56.7 g or 87.2 percent.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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